Math Overview — 97% RTP

Big Bad Wolf Saloon — Wild-West Video Slot

Document	Math Overview — 97% RTP
Project	Big Bad Wolf Saloon
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FIELD	DETAIL
Document	Math verification summary & overview — Standard (97% RTP) model
Version / date	1.0 · 31 May 2026
Build reference	git cefc206
This model	Standard — win-magnitude scale 1× vs the canonical pays/awards
Companion model	The game also ships a Lean 85% RTP model (same reels & feature; win sizes scaled). It has its own .xlsx/.docx.
Companion spreadsheet	Math_Model_97%_RTP.xlsx
Prepared for	Independent slot mathematician / GLI-style certification review

1. How the math was verified

Bottom line: the "Standard" model returns ~97% of stake to players over the long run, confirmed by a 50,000,000-spin simulation run through the exact code the live game executes.

Method

- **One source of truth.** All pay values, reel strips and bonus rules live in two files (par-sheet.js and mathcore.js). The live game and the verifier import the same files, so what is tested is exactly what players get.
- **Selectable model.** The "Standard" model keeps the SAME reels, trigger rate, hit frequency and volatility shape as the other model and scales every win magnitude (line pays + bonus awards) by 1×. The scale is applied once at load.
- **Monte-Carlo simulation.** The verifier plays 50,000,000 independent spins through the real evaluation logic and measures RTP, hit frequency, trigger rate, volatility and per-symbol return. The statistical error on the total RTP is about ±0.1–0.2% (1σ).
- **Faithful accounting.** A spin that triggers the bonus pays the bonus only (its base line wins are forfeited), exactly as the live game does.
- **Independent cross-checks.** The base game can be enumerated exactly (36,729,792 equally-likely screens); the bonus average cross-checks against the closed-form expected award per house tier; and the in-game MATH and SIM panels run the same math.

Reproduce it yourself

BBW_RTP_MODEL=standard node tools/sim.js 50000000

Two items the reviewer must note

- **RNG.** The reference build uses the standard JavaScript random generator (Math.random). That is NOT a certified RNG and must be replaced with a GLI-certified RNG before real-money play. The math is RNG-agnostic, so swapping in a certified RNG changes none of these numbers.
- **Maximum win.** No maximum-win cap is implemented. The largest win seen was 1,274× stake, but the theoretical maximum is higher. If a jurisdiction requires a published max win or a cap, one must be specified.

2. The game at a glance

METRIC	VALUE	BAND
Theoretical RTP (total)	**96.99%**	**target ~97%**
Base game contribution	**49.37%**	
Bonus feature contribution	**47.62%**	
House edge	**3.01%**	
Hit frequency (any win)	**26.4%**	**~1 in 3.8**
Volatility (σ of per-spin return)	**9.26**	**HIGH**
Bonus trigger rate	**≈ 1 in 179**	**6+ hats**
Average bonus value (given a trigger)	**85.5× stake**	
Largest win in sim	**1,274× stake**	**no cap**
Bonus Buy price	**88× stake**	**≈ value-neutral**
Stake range (total stake/spin)	**0.20 – 50.00**	

Read this as: high-volatility and bonus-driven. About half the return comes from the bonus, and within the base game a large share comes from the expanding wild rather than ordinary line hits.

3. How wins work (243 ways)

There are no fixed paylines. A symbol pays when it lands on consecutive reels starting from the leftmost reel, in any rows. The number of "ways" is the product of how many times the symbol appears on each of those reels.

Win for one symbol = (ways) × (pay for that span) × (stake)

- **Span** = how many consecutive reels from reel 1 the symbol covers (at least 3 to pay).
- **Ways** = product of the symbol's counts on each covered reel (up to 243).
- Several different symbols can each pay on the same spin; their wins add together.

The full pay table for all 15 symbols is in the companion spreadsheet (*Paytable* sheet).

4. The Wolf Wild (expanding)

- The Wolf Wild appears only on the centre reel (reel 3).
- When a wild lands, the whole centre reel turns wild — except any Hard-Hat cells, which are kept so the bonus trigger is never affected.
- A wild substitutes for every paying symbol except the three Hard Hats. It has no pay of its own.

Why this matters: because the wild has no pay, its entire return shows up inside the symbols it helps — chiefly the royals on the centre reel. That is why royals lead the base-game return; most of that is really the wild's contribution.

5. The bonus — Hard Hat Free Spins

- **Trigger:** 6+ Hard Hats anywhere awards 6 free spins (about 1 in 179 spins).
- **Build houses:** each hat marks its cell and upgrades it — straw → stick → brick. Frames stay locked for the whole feature.
- **Retrigger:** landing 3+ hats in a single free spin adds +1 free spin.
- **Line wins:** normal 243-ways wins are still paid during the free spins.
- **Wolf reveal:** when the spins end, the wolf blows every built house down for a cash prize:

HOUSE	AWARD (× STAKE)	JACKPOT	AVG AWARD
Straw	**random 0.46 – 2.44**	**—**	**1.45×**
Stick	**random 2.44 – 9.74**	**4% → 29× (Mini)**	**7.01×**
Brick	**random 7.31 – 44.08**	**4% → 146× (Minor)**	**30.51×**

Mansion Jackpot: building 3 or more brick houses fires a Mansion award of stake × (24.4 + up to 19.7 × number of bricks). Average bonus value is about 85.5× stake.

Bonus Buy

The player may pay **88× stake** to start the feature immediately, from a trigger drawn from the same distribution as a natural one. At that price the buy returns about 97.1% — the same value as spinning for it.

6. Where the return comes from

COMPONENT	RTP	SHARE OF TOTAL
Base game (line wins + expanding wild)	**49.37%**	**50.9%**
Bonus feature	**47.62%**	**49.1%**
Total	**96.99%**	**100%**

A per-symbol breakdown of the base-game return is in the spreadsheet (RTP by Symbol sheet).

7. What's in this verification package

FILE	WHAT IT CONTAINS
Math_Model_97%_RTP.xlsx	**This model's workbook: summary, read-me, full pay table, reel counts, exact reel strips, landing probabilities, bonus award math, per-symbol RTP, and constants. Totals/probabilities/expectations are live Excel formulas.**
Math_Overview_97%_RTP.docx	**This document.**
(companion Lean 85% RTP files)	**The other RTP model's matching .xlsx and .docx.**
src/math/par-sheet.js	**Canonical par sheet (pays, reel counts, bonus config, RTP_MODELS).**
src/math/mathcore.js	**Canonical evaluation + bonus play-through logic.**
tools/sim.js	**Headless Monte-Carlo verifier. Run "BBW_RTP_MODEL=standard node tools/sim.js" to reproduce every figure.**

Note: if any value in par-sheet.js changes, re-run the verifier and re-issue this package — every figure here is tied to build cefc206.